

A finite-slice reduction for Josefson–Nissenzweig witnesses in $C_p(X, E)$

Candidate partial result for arXiv:2107.03211, Problem 4.8

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Status. Unconditional partial result, likely valid, pending human review. The unrestricted problem remains open after the bounded literature check recorded below.

1 Source problem

Bargetz, Kąkol, and Sobota ask the following in Problem 4.8 on page 7 of arXiv:2107.03211v3 [1].

Let X be a Tychonoff space and E a locally convex space. Assume that $C_p(X, E)$ has the Josefson–Nissenzweig property. Does it follow that $C_p(X)$ has the Josefson–Nissenzweig property or E has the Josefson–Nissenzweig property?

The proof for the case when $C_p(X)$ has the JNP is analogous. □

We do not know whether the converse to Proposition 4.7 holds.

Problem 4.8. *Let X be a Tychonoff space and E a locally convex space. Assume that the space $C_p(X, E)$ has the JNP. Does it follow that $C_p(X)$ has the JNP or E has the JNP?*

Figure 1: Problem 4.8 and its immediate context in the source paper.

2 Definitions and canonical form

For a locally convex space G , a set $B \subseteq G$ is *barrel-bounded* if every barrel in G absorbs B . The topology $\beta^*(G', G)$ is uniform convergence on barrel-bounded subsets of G . The space G has the JNP when there is a sequence in G' which is null for $\sigma(G', G)$ but not null for $\beta^*(G', G)$; see [2, 1].

Set $H = C_p(X, E)$. The source paper recalls the algebraic identification

$$H' = L_p(X) \otimes E'.$$

After grouping equal evaluation points, every $\Phi \in H'$ has the form

$$\Phi(g) = \sum_{x \in A} \eta_x(g(x)), \quad A \subseteq X \text{ finite}, \quad \eta_x \in E'. \quad (1)$$

The nonzero coefficients in this form are unique. Indeed, a Tychonoff space admits continuous interpolation on each finite subset: for distinct $x_1, \dots, x_m$ there are $h_j \in C(X)$ with $h_j(x_k) = \delta_{jk}$.

3 Proof intuition

A finite slice lets us separate the tensor variables. If only finitely many points of X occur, finite interpolation extracts a weak-star null sequence in E' at each point. If only finitely many coefficient directions in E' occur, a biorthogonal family in E extracts finitely many weak-star null sequences in $L_p(X)$. Evaluation and scalarization are continuous linear maps, so they carry a barrel-bounded witnessing set to barrel-bounded sets. If every extracted component converged uniformly there, their finite sum would also converge uniformly, contradicting the original JNP witness.

Thus neither kind of one-sided finite confinement can produce a genuinely mixed counterexample to Problem 4.8. The remaining case is precisely the two-sided escape regime: infinitely many evaluation points and an infinite-dimensional coefficient span.

4 Finite-slice theorem

Lemma 1 (barrel-bounded images). *If $T : G \rightarrow F$ is continuous linear and $B \subseteq G$ is barrel-bounded, then $T(B)$ is barrel-bounded in F .*

Proof. For every barrel $V \subseteq F$, the inverse image $T^{-1}(V)$ is closed, absolutely convex, and absorbing in G , hence is a barrel. It absorbs B , so V absorbs $T(B)$. $\square$

Theorem 1 (finite-slice reduction). *Let X be Tychonoff, let E be locally convex, and put $H = C_p(X, E)$. Suppose $(\Phi_n) \subseteq H'$ is weak-star null but not $\beta^*(H', H)$ -null. Write*

$$\Phi_n(g) = \sum_{x \in A_n} \eta_{n,x}(g(x))$$

in grouped form. Then:

- (i) *if $\bigcup_n A_n$ is finite, then E has the JNP;*
- (ii) *if $\text{span}\{\eta_{n,x} : n \in \mathbb{N}, x \in A_n\}$ is finite-dimensional, then $C_p(X)$ has the JNP.*

Consequently, if neither $C_p(X)$ nor E has the JNP, every JNP witness in $C_p(X, E)'$ must escape every finite evaluation slice and every finite-dimensional coefficient slice.

Proof. Because (Φ_n) is not β^* -null, there is a barrel-bounded set $B \subseteq H$ on which uniform convergence fails.

For (i), let $A = \bigcup_n A_n$ and put $\eta_{n,x} = 0$ when $x \notin A_n$. Fix $x \in A$. Choose $h_x \in C(X)$ which is 1 at x and 0 on $A \setminus \{x\}$. For $e \in E$, the function $g_{x,e}(t) = h_x(t)e$ belongs to H , and

$$\eta_{n,x}(e) = \Phi_n(g_{x,e}) \longrightarrow 0.$$

Thus $(\eta_{n,x})_n$ is weak-star null in E' for every $x \in A$. By Lemma 1, the set $B_x = \{g(x) : g \in B\}$ is barrel-bounded in E . If every $(\eta_{n,x})_n$ were $\beta^*(E', E)$ -null, then

$$\sup_{g \in B} |\Phi_n(g)| \leq \sum_{x \in A} \sup_{e \in B_x} |\eta_{n,x}(e)| \longrightarrow 0,$$

contrary to the choice of B . Hence one component witnesses the JNP of E .

For (ii), let $F \subseteq E'$ be the finite-dimensional span of all coefficients and choose a basis $e'_1, \dots, e'_m$ of F . There are $e_1, \dots, e_m \in E$ with $e'_j(e_k) = \delta_{jk}$. To see this, the map $E \rightarrow \mathbb{K}^m$, $e \mapsto (e'_1(e), \dots, e'_m(e))$, is onto; otherwise a nonzero functional on $\mathbb{K}^m$ annihilating its range would give a linear dependence among the e'_j .

Write

$$\eta_{n,x} = \sum_{j=1}^m a_{n,x,j} e'_j, \quad \mu_{n,j} = \sum_{x \in A_n} a_{n,x,j} \delta_x \in L_p(X).$$

For $f \in C(X)$, apply Φ_n to $t \mapsto f(t)e_j$ to obtain $\mu_{n,j}(f) \rightarrow 0$. Hence $(\mu_{n,j})_n$ is weak-star null in $C_p(X)'$.

The scalarization map $S_j : H \rightarrow C_p(X)$, $S_j(g) = e'_j \circ g$, is continuous linear. Thus $D_j = S_j(B)$ is barrel-bounded by Lemma 1. If every $(\mu_{n,j})_n$ were $\beta^*(C_p(X)', C_p(X))$ -null, then

$$\sup_{g \in B} |\Phi_n(g)| \leq \sum_{j=1}^m \sup_{f \in D_j} |\mu_{n,j}(f)| \rightarrow 0,$$

again a contradiction. One component therefore witnesses the JNP of $C_p(X)$. □

5 Scope, novelty, and verification

Theorem 1 does not settle Problem 4.8: a weak-star null tensor sequence can have both unbounded evaluation support and infinite-dimensional coefficient span. It shows that any counterexample must use both phenomena essentially, rather than merely embedding a witness from one factor in a disguised finite slice.

The proof is algebraic-topological and uses no computation. The main audit points are Lemma 1, finite interpolation on X , and the finite sum estimate. All are explicit above.

A bounded novelty search covered the run's lightweight indexes and exact arXiv queries using the source title, Problem 4.8, $C_p(X, E)$, JNP, JN-sequences, $L_p(X) \otimes E'$, finite support, and finite-dimensional coefficient span. The background paper arXiv:2003.06764 [2] and the latest source version were checked. No exact finite-slice result or later solution of Problem 4.8 was found. Novelty is plausible but not certified.

Human-review recommendation. Review as a likely valid partial reduction. The priority is to verify the barrel-bounded image step and then decide whether known tensor-sequence compactness principles can force one of the two finite-slice alternatives.

References

- [1] C. Bargetz, J. Kąkol, and D. Sobota, *On complemented copies of the space c_0 in spaces $C_p(X, E)$* , arXiv:2107.03211v3, 2023.
- [2] T. Banach and S. Gabrielyan, *The Josefson–Nissenzweig property for locally convex spaces*, arXiv:2003.06764, 2020.